

Applied Statistics – July 2025 Written Exam - Impact on Severity of Health Insurance Claim

*Provide a small report that explains and motivates your analysis step by step; include all the relevant graphs and summary information, not the code. **Upload the code separately as a script file.***

The report must not exceed 4 pages and must be uploaded to Classroom as a pdf file.

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An insurance claim is a request to an insurance company for payment related to an accident, illness, damage to property, etc. Within this framework, this work wants to investigate claims of a health insurance company.

Gaining an understanding of how the amount (in euros) of the claims paid by the company is related to their customer features is a key factor in the company's decision-making. Particularly, it will help in deciding who are the customers to target if a marketing campaign should be planned. The less the claim, the more profitable will be the customer for the company.

With that in mind, data related to health insurance claims have been collected by an insurance company.

The dataset, available in the *insurance_data.csv* file, contains insightful information related to insurance claims, giving us an in-depth look into the demographic and geographical patterns of those receiving them. Particularly, information on patient age, gender, BMI (Body Mass Index), blood pressure levels, diabetic status, number of children, smoking status, and residential region are available.

Available variables

age: Customer's age

gender: Customer's gender

bmi: Customer's Body Mass Index

bloodpressure: Customer's Blood Pressure Level

Region: Where the insured person lives (Northeast, Northwest, Southeast, Southwest)

Diabetic: Whether the insured person is diabetic or not.

Children: Number of children of the insured person.

Smoker: Whether the insured person is a smoker or not.

Claim: Amount of the health insurance claim.

Your task is to

- (1) understand which of the available variables are mostly related to the insurance claim, and give suggestions to the company on the profiles of the customers a marketing campaign should point at
- (2) evaluate if there is any significant difference between claims made by males rather than by females, given the effect of the other factors
- (3) evaluate which region (among the 4) is more profitable for the company, while taking into account the effect of the other factors.

To read the file:

```
mydata <-  
read.csv("C:\\Users\\Giovanni\\Documents\\Applied\\Exams\\2023_January\\Insurance_Claim.csv")
```